

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
Semester: July 17- October 2017

Max. Marks: 30

Class: SY

Branch: EXTC

Full name of the course : Digital Logic Design

Duration: 1hr.15 min.

Semester: III

Test 1

Course Code: UETC 303

Q. No	Questions	Marks	CO Mapped
Q.1(a)	Solve the following I) Convert $(10100110100101.10001100)_2$ in to base 16 number $10\ 1001\ 1010\ 0101.\ 1000\ 1100$ $2\ 9\ A\ 5\ .\ 8\ C$ II) Convert $(724.63)_{10} = (?)_4 = (23110)_4$ $(724)_{10} = (23110)_4$ $(0.63)_{10} = (0.220)_4$ III) Subtract $(6.25)_{10}$ from $(10.75)_{10}$ using binary arithmetic $1010.11-----10.75$ $-\ 0110.01-----6.25$ $0100.10-----4.50$ $(0100.10)_2 = (4.50)_{10}$ IV) Perform binary subtraction using 1's complement method. $(43-57)_{10}$ $(57)_{10} - 111001$ $(43)_{10} - 101011 \rightarrow 1's\ C\ of\ 57 \rightarrow 000110$ $43 - 101011$ $- 57\ 000110$ $110001 - Ans \rightarrow (001110) -ve\ Ans$ V) Perform binary subtraction using 2's complement method. $(43-57)_{10}$ $2's\ of\ 57\ 000110 + 1 \rightarrow 000111$ $43\ 101011$ $- 57\ 000111$ 110010 $Ans\ is\ -ve\ and\ is\ 2's\ form$ One mark each with proper procedure.	5	CO1

Q.1 (b)	State and Prove D'Morgans First and Second Theorem. Statements-----2M Equations-----1M Proof -----2M Or With suitable example explain - self-complementary code-----3M with suitable Example - Weighted and non-weighted codes-2M with suitable Example	5	CO1
Q.2	Minimize the following expression and Implement using basic gates and all NAND gates. $F(ABCD) = \sum m(1,2,3,5,9,12,14,15) + \sum d(4,8,11)$ K-Map drawing----- 2M K-map grouping with minimized equation- 3M Implementation using basic gates ----- 3M Implementation using all NAND gates -----2M	10	CO1
Q.3	Implement odd parity checker for a 4 bit data using 8:1 mux and additional gates if required. Truth Table ----- 3M Design using 8:1 mux ----- 4M Implementation using 8:1 Mux with enable, Vcc and Gnd—3M	10	CO2

